

Statistical Inference Leveraging Synthetic Data with Distribution-Free Guarantees

Edgar Dobriban

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November 27, 2025



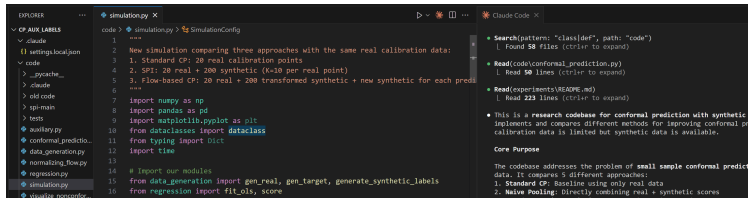
Context

Artificial Intelligence (AI) is emerging as an important technology

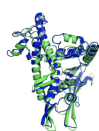
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Success stories: Large language models (ChatGPT, Claude), Image generation (Dall-E, Stable Diffusion, FLUX), Protein structure prediction ...



```
1 """
2 New simulation comparing three approaches with the same real calibration data:
3 1. Standard CP: 20 real calibration points
4 2. SPI: 20 real + 200 synthetic (K=10 per real point)
5 3. Flow-based CP: 20 real + 200 transformed synthetic + new synthetic for each pred
6 """
7 import numpy as np
8 import pandas as pd
9 import matplotlib.pyplot as plt
10 from dataclasses import dataclass
11 from typing import Dict
12 import time
13
14 # Import our modules
15 from data_generation import gen_real, gen_target, generate_synthetic_labels
16 from regression import fit_ols, score
```



T1037 / 6vr4
90.7 GDT
(RNA polymerase domain)



T1049 / 6y4f
93.3 GDT
(adhesin tip)

● Experimental result
● Computational prediction



Solving a Research Problem in Mathematical Statistics with AI Assistance

Edgar Dobriban*

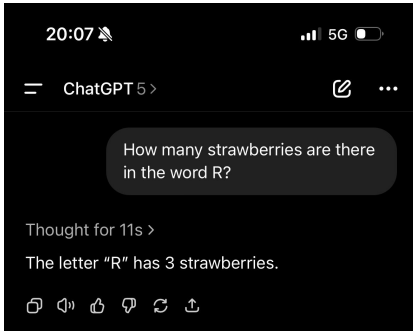
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Abstract

Over the last few months, AI models including large language models have improved greatly. There are now several documented examples where they have helped professional mathematical scientists prove new results, sometimes even helping resolve known open problems. In this short note, we add another example to the list, by documenting how we were able to solve a previously unsolved research problem in robust mathematical statistics with crucial help from GPT-5. Our problem concerns robust density estimation, where the observations are perturbed by Wasserstein-bounded contaminations. In a previous preprint (Chao and Dobriban, 2023, arxiv:2308.01853v2), we have obtained upper and lower bounds on the minimax optimal estimation error; which were, however, not sharp.

Starting in October 2025, making significant use of GPT-5 Pro, we were able to derive the minimax optimal error rate (reported in version 3 of the above arxiv preprint). GPT-5 provided crucial help along the way, including by suggesting calculations that we did not think of, and techniques that were not familiar to us, such as the dynamic Benamou-Brenier formulation, for key steps in the analysis. Working with GPT-5 took a few weeks of effort, and we estimate that it could have taken several months to get the same results otherwise. At the same time, there are still areas where working with GPT-5 was challenging: it sometimes provided incorrect references, and glossed over details that sometimes took days of work to fill in. We outline our workflow and steps taken to mitigate issues. Overall, our work can serve as additional documentation for a new age of human-AI collaborative work in mathematical science.

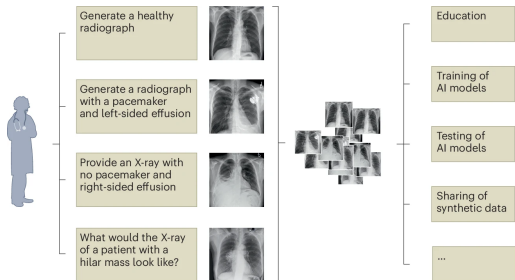
Current AI has problems



The firm found that AI-assisted developers produced three to four times more code than their unassisted peers, but also generated ten times more security issues.



Synthetic Data: Opportunity or Risk?



Truhn et al., 2025

Generative AI produces **synthetic data**

naturegenetics

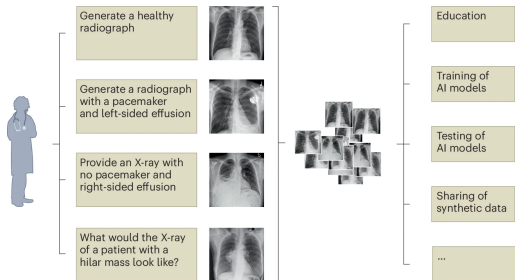
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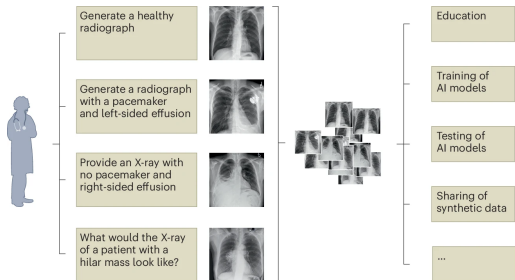
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But synthetic data is imperfect

How can we use synthetic data in statistical inference?

Collaborators



Meshi Bashari



Roy Maor Lotan



Yonghoon Lee



Yaniv Romano

Figure: Collaborators: MB, RML, YR (Technion), YL (Penn).

Overview

GEneral Synthetic-Powered Inference (GESPI)

A **wrapper** method for statistical inference procedures that **safely enhances sample efficiency** by combining synthetic and real data.

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Given a dataset, want to **control a risk measure** (through a loss function ℓ) over the population

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We consider a **risk-controlling method** $\text{Alg} : \mathcal{Z}^\infty \rightarrow \mathcal{A}$ with

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Example: **Hypothesis testing**: $\text{Alg}(\mathcal{D}_n) = \phi_{\mathcal{D}_n} \in \{0, 1\}$ is a test (reject null or not), $\ell(\phi) = \phi$ (no V).

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Example: **Conformal prediction**: $\text{Alg}(\mathcal{D}_n)$ is a prediction set $\hat{\mathcal{C}}_{\mathcal{D}_n} : x \mapsto \hat{\mathcal{C}}_{\mathcal{D}_n}(x)$, $v = (x_{n+1}, y_{n+1})$,
 $\ell(\hat{\mathcal{C}}_{\mathcal{D}_n}, (x, y)) = \mathbb{1} \{y \notin \hat{\mathcal{C}}_{\mathcal{D}_n}(x)\}$.

$$(\text{Coverage error}) = \mathbb{P}\{Y_{n+1} \notin \hat{\mathcal{C}}(X_{n+1})\} = \mathbb{E}\{\mathbb{1}\{Y_{n+1} \notin \hat{\mathcal{C}}(X_{n+1})\}\} \leq \alpha,$$

for all $P \in \{\text{all distributions on } \mathcal{X} \times \mathcal{Y}\}$.

Goal

Goal: Given a synthetic (or more generally, auxiliary) dataset $\tilde{\mathcal{D}}_N$, construct a synthetic-powered procedure $\widetilde{\text{Alg}}(\mathcal{D}_n, \tilde{\mathcal{D}}_N)$ that improves $\text{Alg}(\mathcal{D}_n)$.

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What properties do we desire for $\widetilde{\text{Alg}}(\mathcal{D}_n, \tilde{\mathcal{D}}_N)$?

If the synthetic data is of **high quality**: $\widetilde{\text{Alg}}(\mathcal{D}_n, \tilde{\mathcal{D}}_N)$ behaves **similarly to** $\text{Alg}(\mathcal{D}_{n+N})$.

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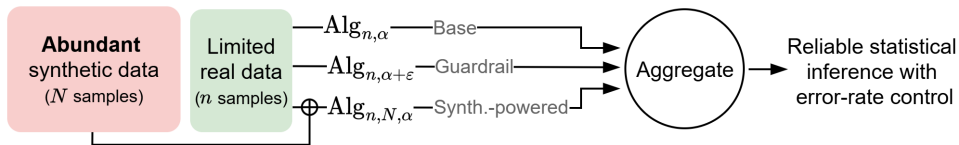
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If the synthetic data is of **low quality**: $\widetilde{\text{Alg}}(\mathcal{D}_n, \tilde{\mathcal{D}}_N)$ does **not rely much** on $\tilde{\mathcal{D}}_N$.

A bird's eye view



Key technical condition required: partially ordered action space

To be robust, we need to be able to **compare actions** in order to choose **the better one**, which has a **lower loss**.

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Condition (informal)

(i) The action space $\mathcal{A}$ is **partially ordered** by $\preceq$, and for any $a_1, a_2 \in \mathcal{A}$ the minimum $a_1 \wedge a_2$ is well defined.

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These are **mild conditions** satisfied in many inference problems.

Example	$\mathcal{A}$	$a_1 \preceq a_2$	$a_1 \wedge a_2$
Confidence intervals	$\mathcal{B}(\mathcal{X})$	$a_1 \supseteq a_2$	$a_1 \cup a_2$
Hypothesis testing	$\{0, 1\}$	$a_1 \leq a_2$	a_1 AND a_2
Predictive inference	$\{g : \mathcal{X} \rightarrow \mathcal{B}(\mathcal{Y})\}$	$a_1(x) \supseteq a_2(x) \ \forall x$	$x \mapsto a_1(x) \cup a_2(x)$

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If the **synthetic data is similar to the real data**, then the first component effectively runs the algorithm with a larger sample size.

However, even **if the synthetic data is dissimilar to the real data**, and the output of the first component is very bad (large according to the order—e.g., small interval), the second guardrail component kicks in to ensure that the overall output is bounded above.

Theoretical guarantees for the one-sided guardrail GESPI procedure

Let P and Q denote the distributions of the real and synthetic data, respectively.

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Theorem (informal)

If the loss is bounded by $c > 0$, the GESPI procedure $\widetilde{\text{Alg}}(\mathcal{D}_n, \tilde{\mathcal{D}}_N)$ satisfies

$$\mathbb{E} \left[\ell(\widetilde{\text{Alg}}(\mathcal{D}_n, \tilde{\mathcal{D}}_N), V) \right] \leq \alpha + \min\{\varepsilon, c \cdot d_{P,Q}\} \quad \text{for all } P, Q \in \mathcal{P},$$

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If $d_{P,Q}$ is large (i.e., low quality synthetic data): the risk is still bounded by $\alpha + \varepsilon$.

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Similar bound on risk (see paper)

GESPI: Examples

1. Prediction sets

$$\hat{C}^{\text{GESPI}}(x) := \underbrace{\hat{C}_{n,\alpha}(x)}_{\text{base prediction set}} \cap \left(\underbrace{\hat{C}_{n,N,\alpha}(x)}_{\text{synthetic-leveraging}} \cup \underbrace{\hat{C}_{n,\alpha+\varepsilon}(x)}_{\text{guardrail}} \right).$$

2. Hypothesis testing

$$\phi^{\text{GESPI}} := \underbrace{\phi_{n,\alpha}}_{\text{base test}} \text{ OR } \left(\underbrace{\tilde{\phi}_{n,N,\alpha}}_{\text{synthetic-leveraging}} \text{ AND } \underbrace{\phi_{n,\alpha+\varepsilon}}_{\text{guardrail}} \right).$$

3. Multiple testing

$$\hat{S}^{\text{GESPI}} := \underbrace{\hat{S}_{n,\alpha}}_{\text{base rejection set}} \cup \left(\underbrace{\hat{S}_{n,N,\alpha}}_{\text{synthetic-leveraging}} \cap \underbrace{\hat{S}_{n,\alpha+\varepsilon}}_{\text{guardrail}} \right).$$

Example 1: Prediction sets—conformal prediction/risk control

Suppose we have a prediction set $\hat{C}_{n,\alpha} = \text{Alg}_\alpha(\mathcal{D}_n) \subset \mathbb{R}$ with a guarantee, e.g.,

1. Conformal prediction (Vovk et al., 1999): $\mathbb{P}\{Y_{n+1} \notin \hat{C}_{n,\alpha}(X_{n+1})\} \leq \alpha.$

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Property: $\hat{C}_{n,\alpha+\varepsilon}(x) \subset \hat{C}^{\text{GESPI}}(x) \subset \hat{C}_{n,\alpha}(x).$

- Always a subset of $\hat{C}_{n,\alpha}$. $\Rightarrow$ width no greater than $\hat{C}_{n,\alpha}$.
- Always contains $\hat{C}_{n,\alpha+\varepsilon}$. $\Rightarrow$ (Miscoverage rate) $\leq \alpha + \varepsilon$.

Empirical illustration A: Image classification

Goal: Construct prediction sets of labels for unlabeled images.

Real data: *ImageNet* (Deng et al., 2009)

Synthetic data: *FLUX* (Black Forest Labs, 2024) generated images



Base algorithm: Class-conditional conformal prediction (Vovk et al., 2005)

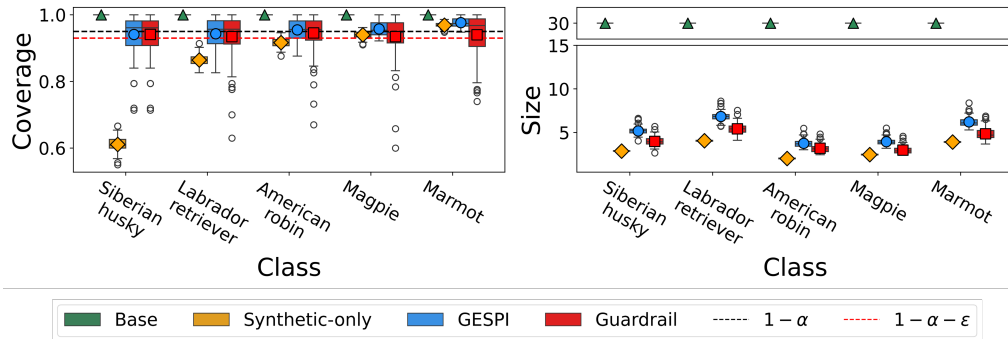
Images: from 30 classes.

- golden retriever, white wolf, beaver, tennis ball, bicycle, . . .

Empirical illustration A: Image classification

Sample sizes: $n = 15$ ($\times 30$ classes), $N = 1000$ ($\times 30$ classes).

Levels: $\alpha = 0.05$, $\varepsilon = 0.02$.

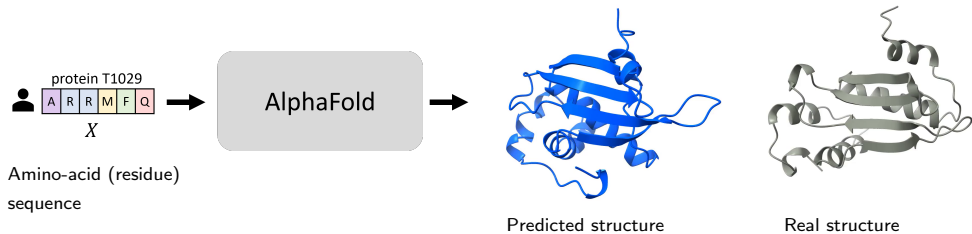


Siberian husky/Labrador retriever: low-quality synthetic data $\Rightarrow$ hits the guardrail.

American robin/Magpie/Marmot: high-quality synthetic data $\Rightarrow$ improved prediction set.

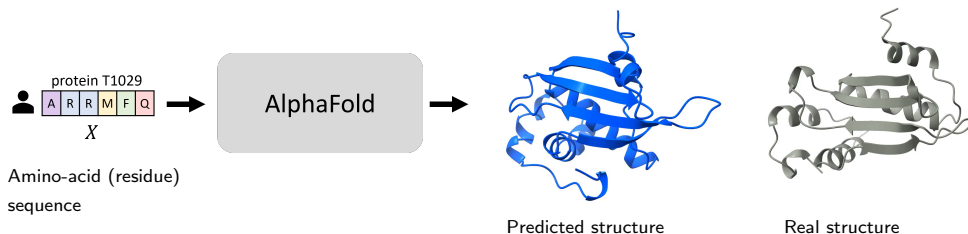
Empirical illustration B: Protein structure prediction

Task: Protein structure prediction with AlphaFold 2.



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Goal: Construct a set $\hat{C}(X)$ of *residues to be abstained on*, satisfying

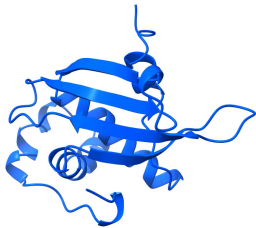
$$\mathbb{E} \left[\frac{1}{|X|} \sum_{i \in X} \mathbb{I}\{\text{err}_i > 3\text{\AA}\} \cdot \mathbb{I}\{i \notin \hat{C}(X)\} \right] \leq \alpha.$$

(measures how many non-abstained residues have large errors)

Base algorithm: Conformal risk control (with per-residue confidence score from AlphaFold).

Empirical illustration B: Protein structure prediction

AlphaFold



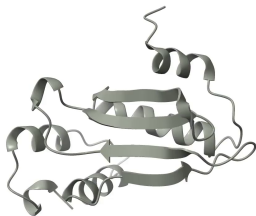
Point prediction

→
conformal risk control



Uncertainty quantification

Real



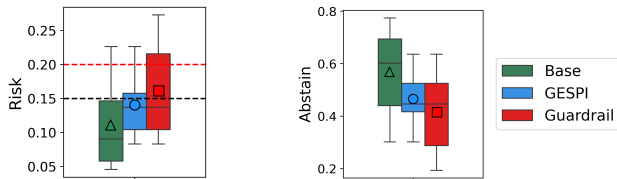
Empirical illustration B: Protein structure prediction

Real data: **CASP14** (community benchmark)

Synthetic data: Multi-sequence alignments (**MSAs**) (key component of AlphaFold 2)

Sample sizes: $n = 10$, $N = 1,000$.

Levels: $\alpha = 0.15$ and $\varepsilon = 0.05$.



The MSA component of AlphaFold 2 can be used beyond its original purpose of improving point predictions.

Example 2: Hypothesis testing

Suppose we have a test $\phi_{n,\alpha} = \text{Alg}_\alpha(\mathcal{D}_n) \in \{0, 1\}$ with type I error control,

$$\mathbb{E}_P[\phi_{n,\alpha}] \leq \alpha, \quad \text{for all null distributions } P.$$

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GESPI testing procedure

$$\phi^{\text{GESPI}} := \underbrace{\phi_{n,\alpha}}_{\text{base test}} \text{ OR } \left(\underbrace{\tilde{\phi}_{n,N,\alpha}}_{\text{synthetic-leveraging}} \text{ AND } \underbrace{\phi_{n,\alpha+\varepsilon}}_{\text{guardrail}} \right).$$

- Always rejects when $\phi_{n,\alpha}$ rejects. $\Rightarrow$ Power at least as high as $\phi_{n,\alpha}$.
- Never rejects when $\phi_{n,\alpha+\varepsilon}$ does not reject. $\Rightarrow$ (Type I error) $\leq \alpha + \varepsilon$.

Empirical illustration C: Testing for the win rate of large reasoning models

We test whether model A outperforms model B on a benchmark:

$$\mathcal{H}_0 : p = 0.5 \text{ versus } \mathcal{H}_1 : p > 0.5,$$

where p is the *win rate of model A over model B*.

Empirical illustration C: Testing for the win rate of large reasoning models

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Goal: Comparing performance on math problems: is a newer but smaller model better than an older but larger one?

Qwen3-1.7B vs DeepSeek-R1-Distill-Qwen-7B.

Base algorithm: Binomial test.

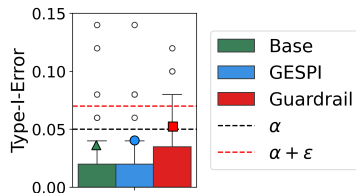
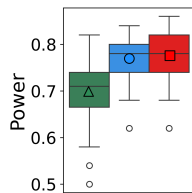
Real dataset: *AIME25* (American Invitational Mathematics Examination)

Synthetic dataset: *Olympiad Bench* (He et al., 2024)

Empirical illustration C: Testing for the win rate of large reasoning models

Sample sizes: $n = 15$, $N = 100$.

Levels: $\alpha = 0.05$ and $\varepsilon = 0.02$.



Under H_1 : higher power.

Under H_0 : Type I error still controlled at level α .

Theory for two-sided GESPI

Tightness assumption for Alg_α

We assume Alg_α is **near-tight** at level α :

$$\alpha - \delta_n \leq \mathbb{E}_{\mathcal{D}_n \sim P, V \sim \mathcal{T}(P)} [\ell(\text{Alg}_\alpha(\mathcal{D}_n), V)] \leq \alpha, \quad \forall P \in \mathcal{P}, n \in \mathbb{N}. \quad (1)$$

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Examples of δ_n :

Normal mean CI: $[\bar{X} - z_{\alpha/2} \sigma / \sqrt{n}, \bar{X} + z_{\alpha/2} \sigma / \sqrt{n}]$: $\delta_n = 0$.

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Conformal risk control (Angelopoulos et al., 2022): $\delta_n = \frac{2c}{n+1}$ (loss bounded by c).

Extended result for two-sided guardrail GESPI

Theorem (extended two-sided guardrail)

If Alg_α satisfies (1) for levels α and $\alpha + \varepsilon$, then for all $P, Q \in \mathcal{P}$,

$$\alpha - \min\{\delta_n, c\tau + \delta_{N+n} + c\tilde{d}_{\ell, \text{Alg}}(P, Q)\} \leq \mathbb{E} \left[\ell(\widetilde{\text{Alg}}(\mathcal{D}_n, \tilde{\mathcal{D}}_N), V) \right] \leq \alpha + \min\{\varepsilon, c\tau + c\tilde{d}_{\ell, \text{Alg}}(P, Q)\},$$

where

$$\tilde{d}_{\ell, \text{Alg}}(P, Q) = d_{\text{TV}}(\tilde{P}_{\ell, \text{Alg}}(P, Q), \tilde{P}_{\ell, \text{Alg}}(Q, Q)),$$

$$\text{and } \tau = 1 - \mathbb{P}_{\mathcal{D}_n \cup \tilde{\mathcal{D}}_N \stackrel{\text{iid}}{\sim} Q} \left\{ \text{Alg}_\alpha(\mathcal{D}_n) \preceq \text{Alg}_\alpha(\mathcal{D}_n \cup \tilde{\mathcal{D}}_N) \preceq \text{Alg}_{\alpha+\varepsilon}(\mathcal{D}_n) \right\}.$$

Example 3: Multiple testing

Given m hypotheses $H_1, \dots, H_m$, consider a **multiple testing procedure** that outputs $\hat{S}$ with k -FWER control:

$$\mathbb{P} \left\{ \sum_{j=1}^m \mathbb{1} \left\{ H_j \text{ is true but } j \in \hat{S} \right\} \geq k \right\} \leq \alpha.$$

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Does not reject hypotheses not rejected by $\hat{S}_{n,\alpha+\varepsilon} \Rightarrow k\text{-FWER} \leq \alpha + \varepsilon$.

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We developed GESPI, a method to leverage synthetic data for statistical inference

- Helps when synthetic data resembles real data

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Pre-print: <https://arxiv.org/abs/2509.20345>

Thanks!



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Problem: Robust density estimation under Wasserstein contamination

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Human-only (Feb. 2021 - Aug. 2023):

$$\max \left\{ n^{-2s/(2s+1)}, \varepsilon^{2s/(s+1/2)} \right\} \lesssim \mathcal{M}(\varepsilon, s, 1, 2, 2) \lesssim \max \left\{ n^{-2s/(2s+1)}, \varepsilon^{2s/(s+2)} \right\}.$$

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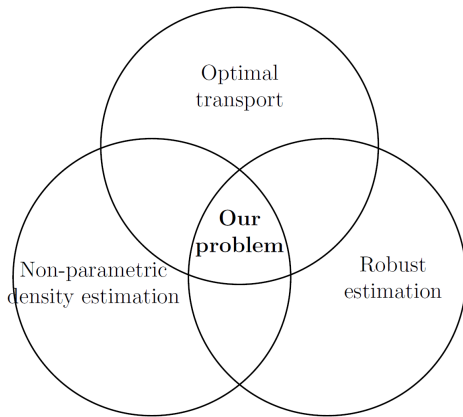
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AI-assisted (Oct. - Nov. 2025):

$$\mathcal{M}(\varepsilon, s, p, q, r) \asymp \max \left\{ n^{-\frac{2s}{2s+p}}, \varepsilon^{\frac{2s}{s+1+p/r}} \right\}.$$

In particular, $\mathcal{M}(\varepsilon, s, 1, 2, 2) \asymp \max \left\{ n^{-\frac{2s}{2s+1}}, \varepsilon^{\frac{2s}{s+3/2}} \right\}.$

Solving a Statistics Research Problem with AI Help (ctd)



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In our **lower bound**, we have adapted the modulus of continuity approach, by starting with a normal density $f = \varphi_\sigma$ with sufficiently small variance $\sigma^2 > 0$, and perturbing it with a mean zero function to obtain a contaminated density $\tilde{f}$. Then, we upper bounded $W_{2,2}(f, \tilde{f}) \leq \varepsilon$ (via the Talagrand transportation cost inequality, see Talagrand (1996); Otto and Villani (2000)) and lower bounded $|f(x_0) - \tilde{f}(x_0)|$, which—by the modulus of continuity approach—lead to a lower bound on the estimation error.

Solving a Statistics Research Problem with AI Help (ctd)

GPT-5 Suggests Dynamic Benamou-Brenier for Improved Lower Bound (Extract)

User: [Prompt includes LaTeX draft of paper and a query: "Can you go through this proof and see if you can improve either the upper bound and the lower bound until they match?"]

GPT: [...]

Lemma [...] scales like $W_2^2 \lesssim D^2 \int g_+$, i.e. **it is linear in the bump amplitude τ while the correct small-amplitude scaling for W_2 in one dimension is quadratic in τ .**

[...] We state and prove the replacement for Lemma [...].

Lemma (Refined local W_2 bound via the dynamic formulation)

Let q be a probability density on $\mathbb{R}$ and let $I \subset \mathbb{R}$ be a finite interval. Let $g \in L^1(\mathbb{R})$ satisfy $\text{supp}(g) \subset I$ and $\int g = 0$. Write $G(x) = \int_{-\infty}^x g(t) dt$. Suppose there exists $c_I > 0$ such that $q(x) \geq c_I$ on I and $|g|_\infty \leq c_I/2$. Then $q + g$ is a density and

$$W_2^2(q + g, q) \leq 2c_I \int_{\mathbb{R}} G(x)^2 dx.$$

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Proof.

Consider the path of densities $\rho_t = q + tg$ for $t \in [0, 1]$. On I we have $\rho_t \geq q - |g|_\infty \geq c_I/2$, while off I one has $\rho_t = q > 0$. Define $v_t(x) = -G(x)/\rho_t(x)$. Then $\rho_t v_t = -G$ is independent of t , hence $\partial_x(\rho_t v_t) = -\partial_x G = -g$ and $\partial_t \rho_t = g$; thus $\partial_t \rho_t + \partial_x(\rho_t v_t) = 0$. Applying the Benamou–Brenier dynamic formulation gives

$$W_2^2(q + g, q) \leq \int_0^1 \int_{\mathbb{R}} |v_t(x)|^2 \rho_t(x) dx dt = \int_0^1 \int_{\mathbb{R}} G(x)^2 dx dt = \int_{\mathbb{R}} G(x)^2 dx.$$

Since $\rho_t(x) \geq c_I/2$ on I and $\rho_t = q$ off I , the bound sharpens to

$$W_2^2(q + g, q) \leq 2c_I \int_{\mathbb{R}} G(x)^2 dx.$$

